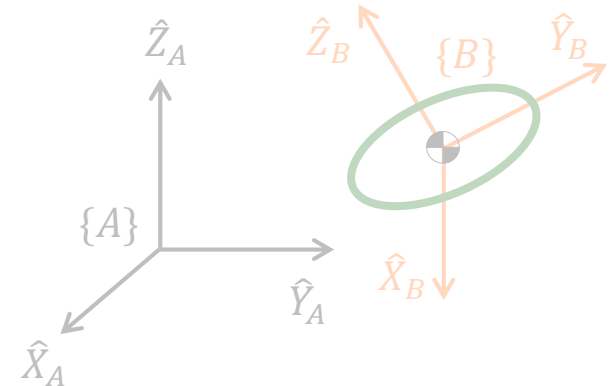


Rotation Matrix與轉角

□ Rotation matrix 的三種用法

- ◆ 描述一個frame(相對於另一個frame)的姿態

$${}^A_B R = \begin{bmatrix} | & | & | \\ {}^A\hat{X}_B & {}^A\hat{Y}_B & {}^A\hat{Z}_B \\ | & | & | \end{bmatrix}$$

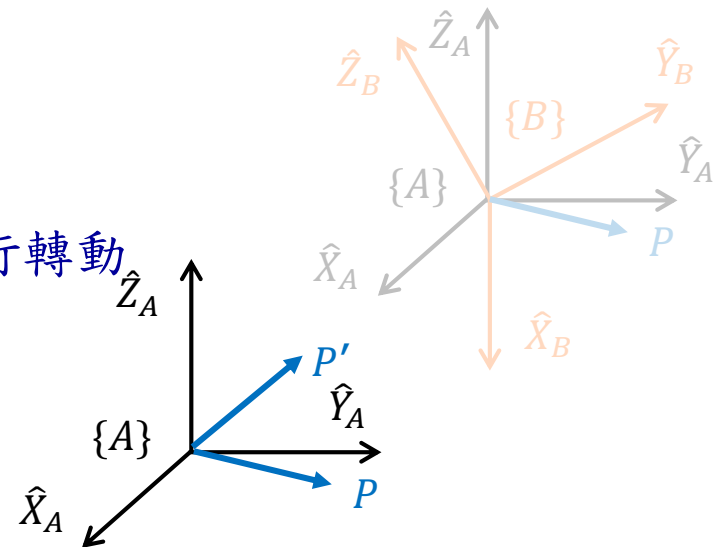


- ◆ 將point由某一個frame的表達換到另一個和此frame僅有相對轉動的frame來表達

$${}^A P = {}^A_B R {}^B P$$

- ◆ 將point(vector)在同一個frame中進行轉動

$${}^A P' = R(\theta) {}^A P$$



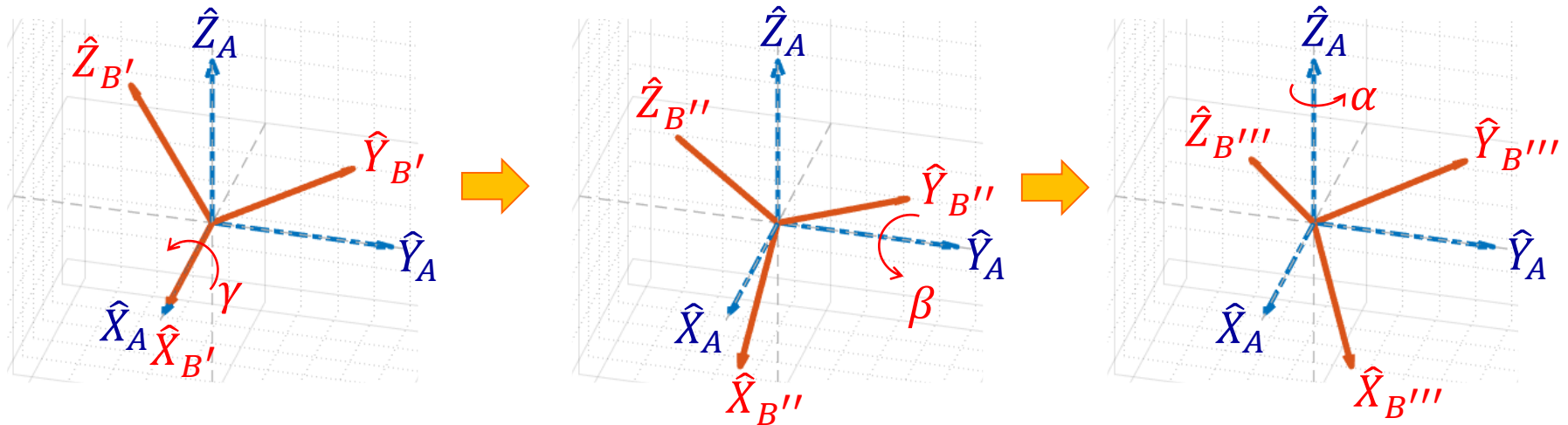


Rotation Matrix與轉角

- 空間中的Rotation是3 DOFs，那要如何把一般rotation matrix所表達的姿態，拆解成3次旋轉角度，以對應到3個 DOFs？
- 拆解成「三次旋轉連乘」所需注意事項
 - ◆ Rotation不是commutable，所以多次旋轉的先後順序需要明確定義
 - ◆ 旋轉轉軸也需要明確定義。是對「固定不動」的轉軸旋轉？或是對「轉動的frame當下所在」的轉軸旋轉？
- 兩個拆解方式
 - ◆ 對方向「固定不動」的轉軸旋轉：Fixed angles
 - ◆ 對「轉動的frame當下所在」的轉軸方向旋轉：Euler angles

Fixed Angles -1

□ X-Y-Z Fixed Angles – 由angles推算R

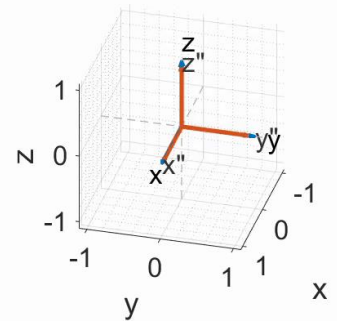


$${}^A_B R_{XYZ}(\gamma, \beta, \alpha) = R_Z(\alpha)R_Y(\beta)R_X(\gamma) \quad v' = {}^A_B R v = R_3 R_2 R_1 v$$

先轉的放「後面」：以operator來想，對某一個向量，
「以同一個座標為基準」，進行轉動或移動的操作

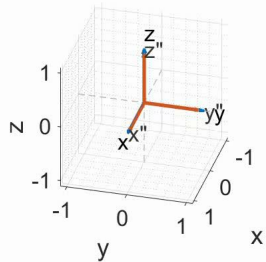
$$= \begin{bmatrix} c\alpha & -s\alpha & 0 \\ s\alpha & c\alpha & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} c\beta & 0 & s\beta \\ 0 & 1 & 0 \\ -s\beta & 0 & c\beta \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & c\gamma & -s\gamma \\ 0 & s\gamma & c\gamma \end{bmatrix}$$

$$= \begin{bmatrix} cac\beta & cas\beta s\gamma - sac\gamma & cas\beta c\gamma + sas\gamma \\ sac\beta & sas\beta s\gamma + cac\gamma & sas\beta c\gamma - cas\gamma \\ -s\beta & c\beta s\gamma & c\beta c\gamma \end{bmatrix}$$



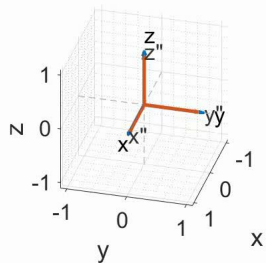
Fixed Angles -2

- Ex: 以Fixed Angles旋轉：「先對X軸旋轉60度，後對Y軸旋轉30度」和「先對Y軸旋轉30度，後對X軸旋轉60度」各自的 ${}^A_B R$ 分別是？



先對X轉60度，再對Y轉30度

$${}^A_B R_{XYZ}(\gamma, \beta, \alpha) = R_Z(0)R_Y(30)R_X(60) = \begin{bmatrix} 0.866 & 0.433 & 0.25 \\ 0 & 0.5 & -0.866 \\ -0.5 & 0.75 & 0.433 \end{bmatrix}$$



先對Y轉30度，再對X轉60度

$${}^A_B R_{XYZ}(\gamma, \beta, \alpha) = R_Z(0)R_X(60)R_Y(30) = \begin{bmatrix} 0.866 & 0 & 0.5 \\ 0.433 & 0.5 & -0.75 \\ -0.25 & 0.866 & 0.433 \end{bmatrix}$$

Fixed Angles -3

□ X-Y-Z Fixed Angles – 由 R 推算 angles

$${}^A_B R_{XYZ}(\gamma, \beta, \alpha) = \begin{bmatrix} c\alpha c\beta & c\alpha s\beta s\gamma - s\alpha c\gamma & c\alpha s\beta c\gamma + s\alpha s\gamma \\ s\alpha c\beta & s\alpha s\beta s\gamma + c\alpha c\gamma & s\alpha s\beta c\gamma - c\alpha s\gamma \\ -s\beta & c\beta s\gamma & c\beta c\gamma \end{bmatrix} = \begin{bmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \end{bmatrix}$$

If $\beta \neq 90^\circ$

$$\beta = \text{Atan2}(-r_{31}, \sqrt{r_{11}^2 + r_{21}^2})$$

$$\alpha = \text{Atan2}(r_{21}/c\beta, r_{11}/c\beta)$$

$$\gamma = \text{Atan2}(r_{32}/c\beta, r_{33}/c\beta)$$

$$-90^\circ \leq \beta \leq 90^\circ$$

Single solution

If $\beta = 90^\circ$

$$\alpha = 0^\circ$$

$$\gamma = \text{Atan2}(r_{12}, r_{22})$$

If $\beta = -90^\circ$

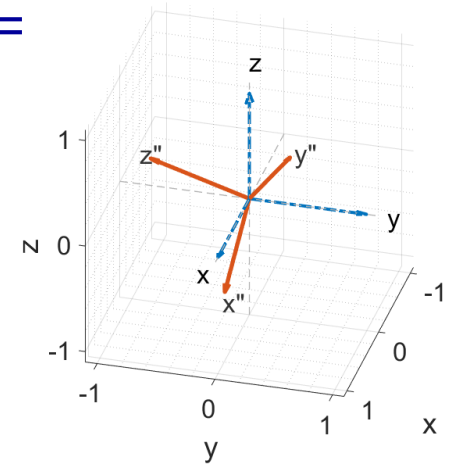
$$\alpha = 0^\circ$$

$$\gamma = -\text{Atan2}(r_{12}, r_{22})$$

Fixed Angles -4

□ Ex: 以X-Y-Z Fixed Angles方法，反算 $R =$

$$\begin{bmatrix} 0.866 & 0.433 & 0.25 \\ 0 & 0.5 & -0.866 \\ -0.5 & 0.75 & 0.433 \end{bmatrix} \text{的angles}$$



$$\beta = \text{Atan2} \left(-r_{31}, \sqrt{r_{11}^2 + r_{21}^2} \right) = \text{Atan2} \left(-(-0.5), \sqrt{0.866^2 + 0^2} \right) = 30^\circ$$

$$\alpha = \text{Atan2} \left(\frac{r_{21}}{c\beta}, \frac{r_{11}}{c\beta} \right) = \text{Atan2} \left(\frac{0}{\cos 30}, \frac{0.866}{\cos 30} \right) = 0^\circ$$

$$\gamma = \text{Atan2} \left(\frac{r_{32}}{c\beta}, \frac{r_{33}}{c\beta} \right) = \text{Atan2} \left(\frac{0.75}{\cos 30}, \frac{0.433}{\cos 30} \right) = 60^\circ$$



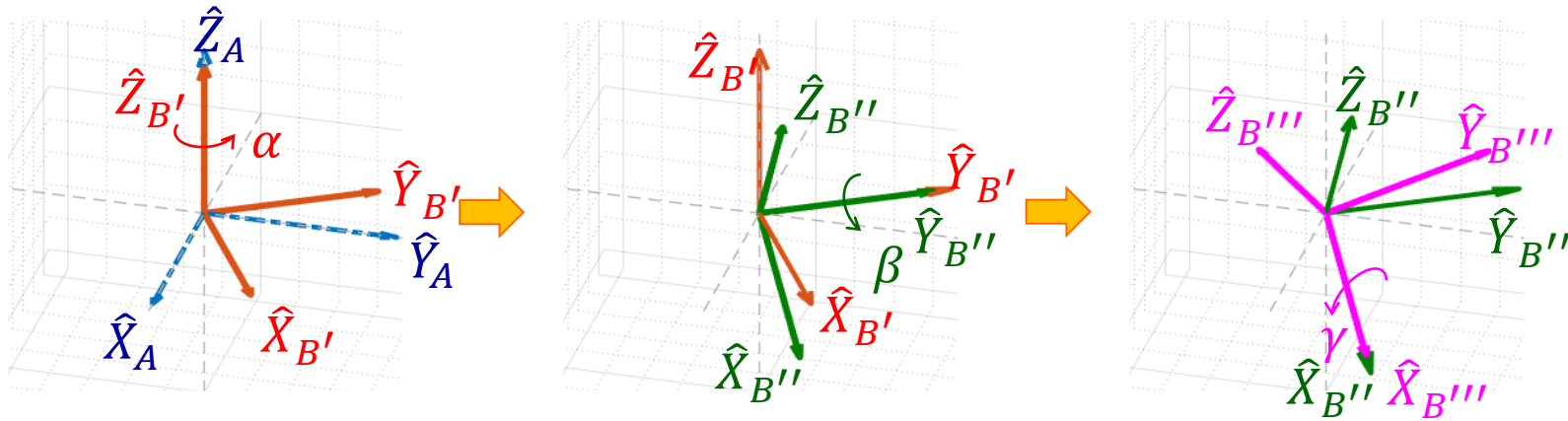
$$R_Z(0)R_Y(30)R_X(60)$$

先對X轉60度，再對Y轉30度

和Fixed Angles -2的結果相同

Euler Angles -1

□ Z-Y-X Euler Angles - 由angles推算R



$${}^A_B R_{Z'Y'X'}(\alpha, \beta, \gamma) = {}_{B'}^{B''} R {}_{B''}^{B'''} R = R_{Z'}(\alpha) R_{Y'}(\beta) R_{X'}(\gamma)$$

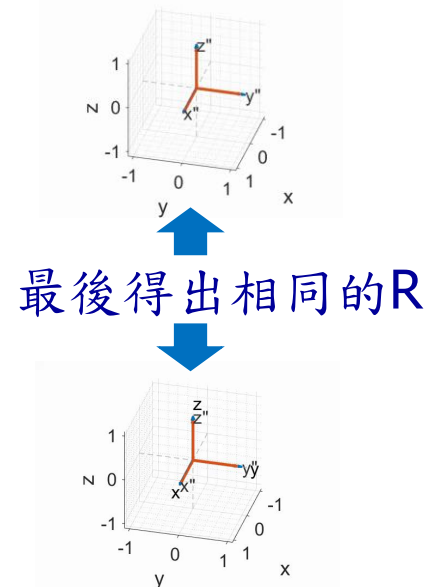
先轉的放「前面」：以mapping來想，對某一個向量，從最後一個frame「逐漸轉動或移動」來回到第一個frame

$${}^A P = {}^A_B R {}^B P = R_1 R_2 R_3 {}^B P$$

$$= \begin{bmatrix} c\alpha & -s\alpha & 0 \\ s\alpha & c\alpha & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} c\beta & 0 & s\beta \\ 0 & 1 & 0 \\ -s\beta & 0 & c\beta \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & c\gamma & -s\gamma \\ 0 & s\gamma & c\gamma \end{bmatrix}$$

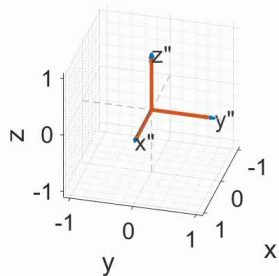
$$= R_Z(\alpha) R_Y(\beta) R_X(\gamma) = {}^A_B R_{XYZ}(\gamma, \beta, \alpha)$$

和X-Y-Z Fixed angle得到一樣的R



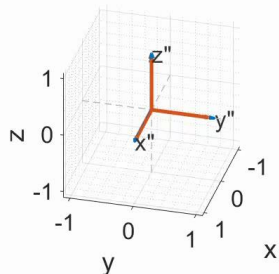
Euler Angles -2

- Ex: 以Euler Angles旋轉：「先對X軸旋轉60度，後對Y軸旋轉30度」和「先對Y軸旋轉30度，後對X軸旋轉60度」各自的 ${}^A_B R$ 分別是？



先對X轉60度，再對Y轉30度

$${}^A_B R_{X'Y'Z'}(\gamma, \beta, \alpha) = R_{X'}(60)R_{Y'}(30) = \begin{bmatrix} 0.866 & 0 & 0.5 \\ 0.433 & 0.5 & -0.75 \\ -0.25 & 0.866 & 0.433 \end{bmatrix}$$

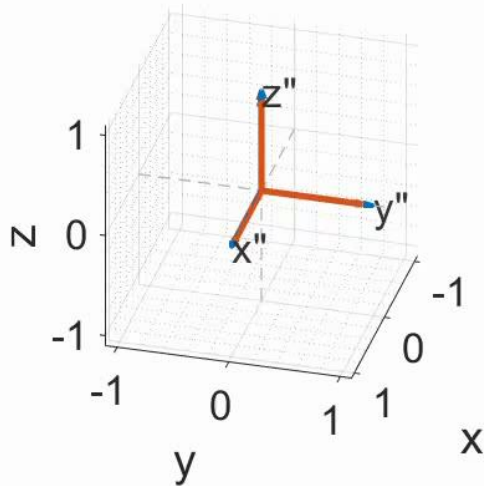


先對Y轉30度，再對X轉60度

$${}^A_B R_{X'Y'Z'}(\gamma, \beta, \alpha) = R_{Y'}(30)R_{X'}(60) = \begin{bmatrix} 0.866 & 0.433 & 0.25 \\ 0 & 0.5 & -0.866 \\ -0.5 & 0.75 & 0.433 \end{bmatrix}$$

Euler Angles -3

- Ex: Euler(Y30, X60) v.s. Fixed(X60, Y30)

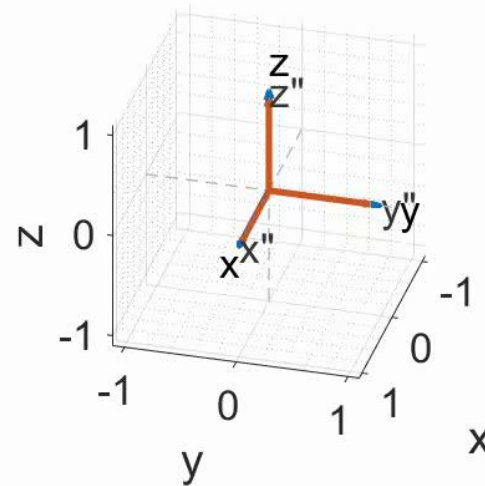


Euler Angles:

先對Y轉30度，再對X轉60度

$$\begin{aligned}
 & {}^A_B R_{x''y''z''}(\gamma, \beta, \alpha) \\
 &= R_{y'}(30)R_{x'}(60) \\
 &= \begin{bmatrix} 0.866 & 0.433 & 0.25 \\ 0 & 0.5 & -0.866 \\ -0.5 & 0.75 & 0.433 \end{bmatrix}
 \end{aligned}$$

=



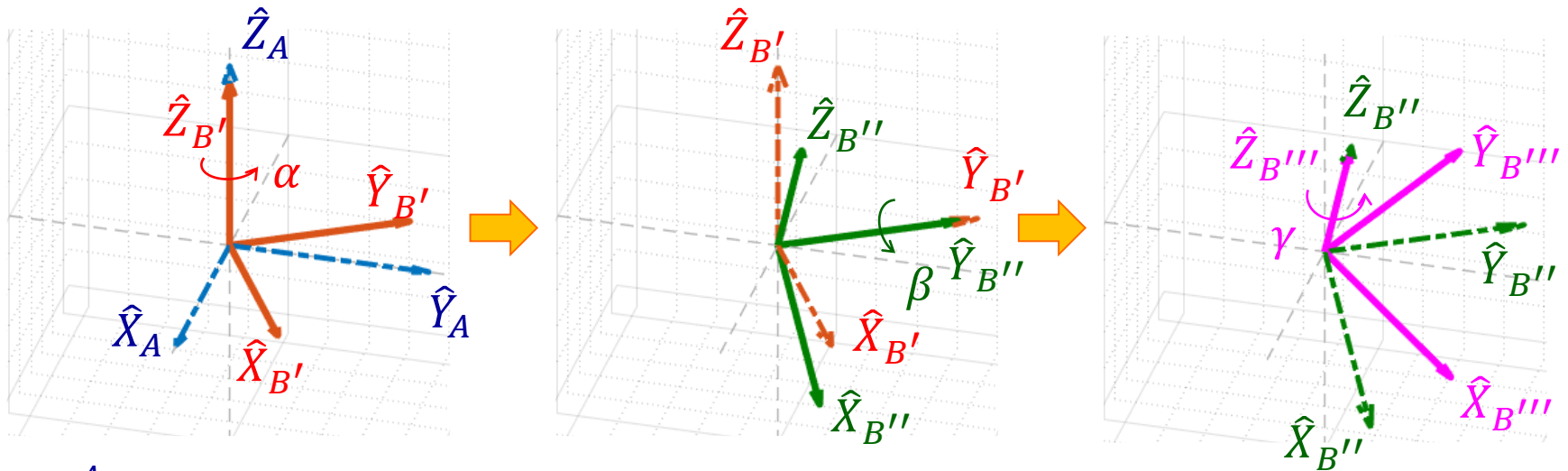
Fixed Angles:

先對X轉60度，再對Y轉30度

$$\begin{aligned}
 & {}^A_B R_{x''y''z''}(\gamma, \beta, \alpha) \\
 &= R_Y(30)R_X(60) \\
 &= \begin{bmatrix} 0.866 & 0.433 & 0.25 \\ 0 & 0.5 & -0.866 \\ -0.5 & 0.75 & 0.433 \end{bmatrix}
 \end{aligned}$$

Euler Angles -4

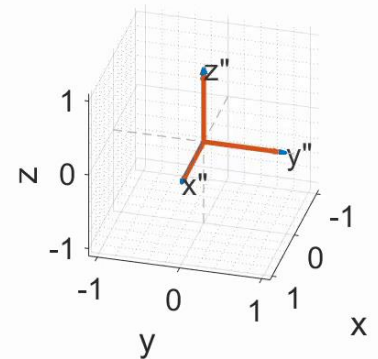
□ Z-Y-Z Euler Angles - 由angles推算R



$${}^A_B R_{Z'Y'Z'}(\alpha, \beta, \gamma) = R_{Z'}(\alpha) R_{Y'}(\beta) R_{Z'}(\gamma)$$

先轉的放「前面」

$$= \begin{bmatrix} c\alpha c\beta c\gamma - s\alpha s\gamma & -c\alpha c\beta s\gamma - s\alpha c\gamma & c\alpha s\beta \\ s\alpha c\beta c\gamma + c\alpha s\gamma & -s\alpha c\beta s\gamma + c\alpha c\gamma & s\alpha s\beta \\ -s\beta c\gamma & s\beta s\gamma & c\beta \end{bmatrix}$$



Euler Angles -5

□ Z-Y-Z Euler Angles - 由 R 推算 angles

$${}^A_B R_{Z'Y'Z'}(\alpha, \beta, \gamma) = \begin{bmatrix} c\alpha c\beta c\gamma - s\alpha s\gamma & -c\alpha c\beta s\gamma - s\alpha c\gamma & c\alpha s\beta \\ s\alpha c\beta c\gamma + c\alpha s\gamma & -s\alpha c\beta s\gamma + c\alpha c\gamma & s\alpha s\beta \\ -s\beta c\gamma & s\beta s\gamma & c\beta \end{bmatrix} = \begin{bmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \end{bmatrix}$$

If $\beta \neq 0^\circ$

$$\beta = \text{Atan2}(\sqrt{r_{31}^2 + r_{32}^2}, r_{33})$$

$$\alpha = \text{Atan2}(r_{23}/s\beta, r_{13}/s\beta)$$

$$\gamma = \text{Atan2}(r_{32}/s\beta, -r_{31}/s\beta)$$

If $\beta = 0^\circ$

$$\alpha = 0^\circ$$

$$\gamma = \text{Atan2}(-r_{12}, r_{11})$$

If $\beta = 180^\circ$

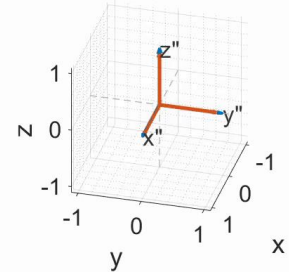
$$\alpha = 0^\circ$$

$$\gamma = \text{Atan2}(r_{12}, -r_{11})$$

Euler Angles -6

- Ex: Revisit Euler Angles-2的範例

$${}^A_B R_{X'Y'Z'}(60,30,0) = R_{X'}(60)R_{Y'}(30) = \begin{bmatrix} 0.866 & 0 & 0.5 \\ 0.433 & 0.5 & -0.75 \\ -0.25 & 0.866 & 0.433 \end{bmatrix}$$



- 若以ZYZ的公式反算，Euler Angles 為何？

$$R_{X'}(60)R_{Y'}(30)$$

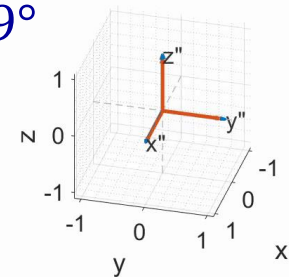
$$\beta = \text{Atan2}\left(\sqrt{r_{31}^2 + r_{32}^2}, r_{33}\right) = \text{Atan2}\left(\sqrt{(-0.25)^2 + 0.866^2}, 0.433\right) = 64.3^\circ$$

$$\alpha = \text{Atan2}\left(\frac{r_{23}}{s\beta}, \frac{r_{13}}{s\beta}\right) = \text{Atan2}\left(\frac{-0.75}{s\beta}, \frac{0.5}{s\beta}\right) = -56.3^\circ$$

$$\gamma = \text{Atan2}(r_{32}/s\beta, -r_{31}/s\beta) = \text{Atan2}(0.866/s\beta, 0.25/s\beta) = 73.9^\circ$$

➡ $R_{Z'}(-56.3)R_{Y'}(64.3)R_{Z'}(73.9)$

先對Z轉 -56.3° ,對Y轉 64.3° ,最後對Z轉 73.9°



$$R_{Z'}(-56.3)R_{Y'}(64.3)R_{Z'}(73.9)$$

Rotation Matrix與轉角 小結

□ Euler/Fixed angles

- ◆ 12種 Euler angles 和 12種 fixed angles
- ◆ 存在Duality – 共12種對principal axes連三次轉動的拆解方法

□ Angle-axis表達法



Unit vector裡2個參數，轉角1個參數，
也為3 DOFs

□ Quaternion表達法

$$\begin{aligned} q &= \epsilon_4 + \epsilon_1 \hat{i} + \epsilon_2 \hat{j} + \epsilon_3 \hat{k} \\ &= \cos \frac{\theta}{2} + \sin \frac{\theta}{2} (k_x \hat{i} + k_y \hat{j} + k_z \hat{k}) \end{aligned}$$

note $\epsilon_1^2 + \epsilon_2^2 + \epsilon_3^2 + \epsilon_4^2 = 1$

4個參數+1個限制條件，也為3 DOFs

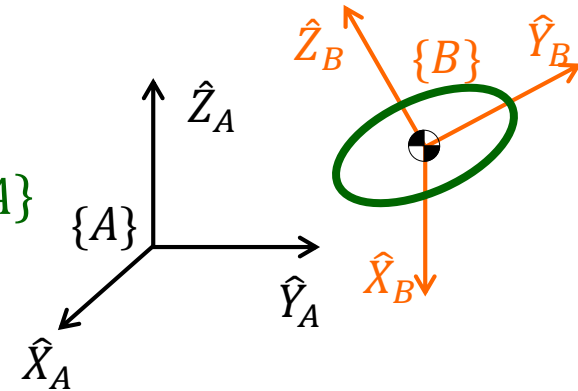


剛體狀態的表達 -1

- 「導讀-3」的問題：該如何整合表達剛體的狀態？
- $\Rightarrow$ 在剛體(Rigid body)上建立frame，常建立在質心上

- ◆ 移動：由body frame 的原點位置判定

$${}^A P_{B\ org} = \begin{bmatrix} P_{B\ x} \\ P_{B\ y} \\ P_{B\ z} \end{bmatrix} = \text{origin of } \{B\} \text{ represented in } \{A\}$$



- ◆ 轉動：由body frame的姿態判定

$${}^A R_B = \begin{bmatrix} \hat{X}_B \cdot \hat{X}_A & \hat{Y}_B \cdot \hat{X}_A & \hat{Z}_B \cdot \hat{X}_A \\ \hat{X}_B \cdot \hat{Y}_A & \hat{Y}_B \cdot \hat{Y}_A & \hat{Z}_B \cdot \hat{Y}_A \\ \hat{X}_B \cdot \hat{Z}_A & \hat{Y}_B \cdot \hat{Z}_A & \hat{Z}_B \cdot \hat{Z}_A \end{bmatrix} = \begin{bmatrix} | & | & | \\ {}^A \hat{X}_B & {}^A \hat{Y}_B & {}^A \hat{Z}_B \\ | & | & | \end{bmatrix}$$

- ◆ 彙整後：

$$\{B\} = \{{}_B^A R, {}^A P_{B\ org}\}$$

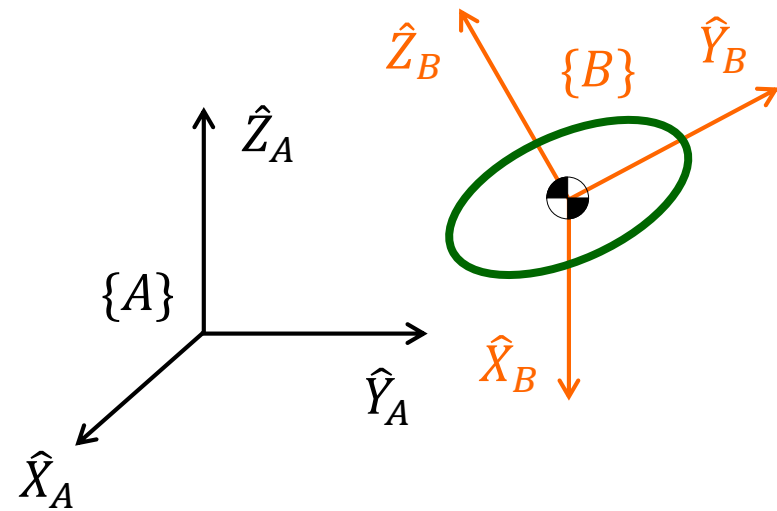
但無法進行量化計算

剛體狀態的表達 -2

□ 如何將移動和轉動整合在一起描述？

⇒ Homogeneous transformation matrix

$$\begin{bmatrix} {}^A R_B & {}^A P_{B \text{ org}} \\ 0 & 1 \end{bmatrix}_{4 \times 4}$$
$$= \begin{bmatrix} | & | & | & | \\ {}^A \hat{X}_B & {}^A \hat{Y}_B & {}^A \hat{Z}_B & {}^A P_{B \text{ org}} \\ | & | & | & | \\ \hline 0 & 0 & 0 & 1 \end{bmatrix}$$
$$= {}^A_B T$$



Mapping -1

□ 以 Mapping，轉換向量（或點）之座標系的方式來確認 ${}^A_B T$

運算之正確性

◆ 僅有 移動

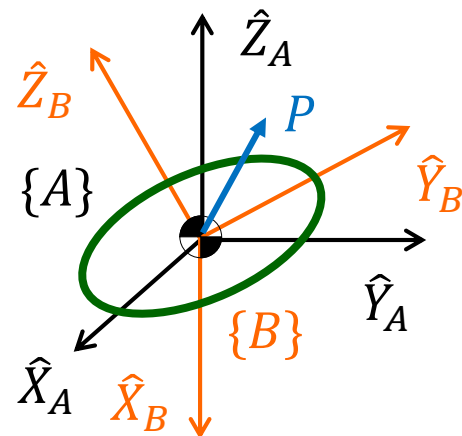
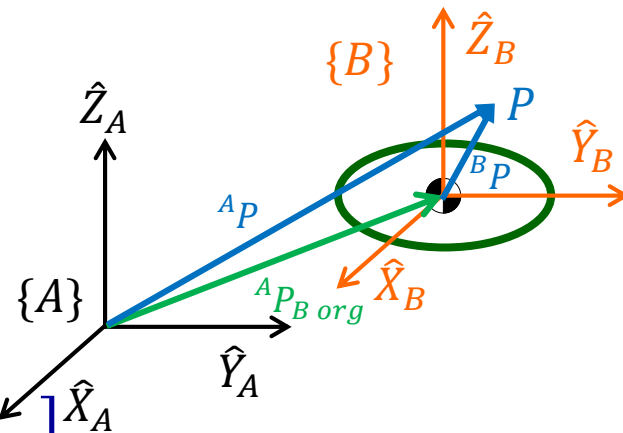
$${}^A P_{3 \times 1} = {}^B P_{3 \times 1} + {}^A P_{B \text{ org}}_{3 \times 1}$$

$$\begin{bmatrix} {}^A P \\ 1 \end{bmatrix} = \begin{bmatrix} I_{3 \times 3} & {}^A P_{B \text{ org}} \\ 0 & 1 \end{bmatrix} \begin{bmatrix} {}^B P \\ 1 \end{bmatrix} = \begin{bmatrix} {}^B P + {}^A P_{B \text{ org}} \\ 1 \end{bmatrix}$$

◆ 僅有 轉動

$${}^A P_{3 \times 1} = {}^A_B R_{3 \times 3} {}^B P_{3 \times 1}$$

$$\begin{bmatrix} {}^A P \\ 1 \end{bmatrix} = \begin{bmatrix} {}^A_B R & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} {}^B P \\ 1 \end{bmatrix} = \begin{bmatrix} {}^A_B R {}^B P \\ 1 \end{bmatrix}$$



Mapping -2

◆ 移動和轉動複合

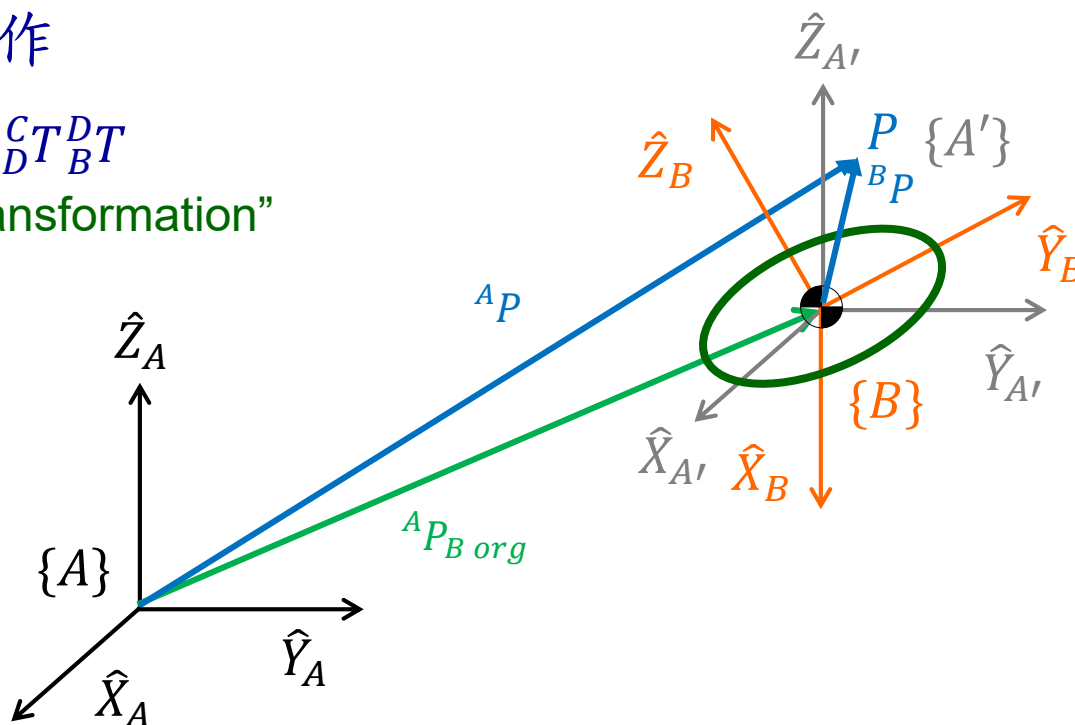
$${}^A P_{3 \times 1} = {}^A R_B {}^B P_{3 \times 1} + {}^A P_{B \text{ org } 3 \times 1}$$

$$\begin{bmatrix} {}^A P \\ 1 \end{bmatrix} = \begin{bmatrix} {}^A R_B & {}^A P_{B \text{ org }} \\ 0 & 1 \end{bmatrix} \begin{bmatrix} {}^B P \\ 1 \end{bmatrix} = \begin{bmatrix} {}^A R_B {}^B P + {}^A P_{B \text{ org }} \\ 1 \end{bmatrix}$$

□ 可連續操作

$${}^A T = {}^A T_C {}^C T_D {}^D T_B T$$

“sequential transformation”

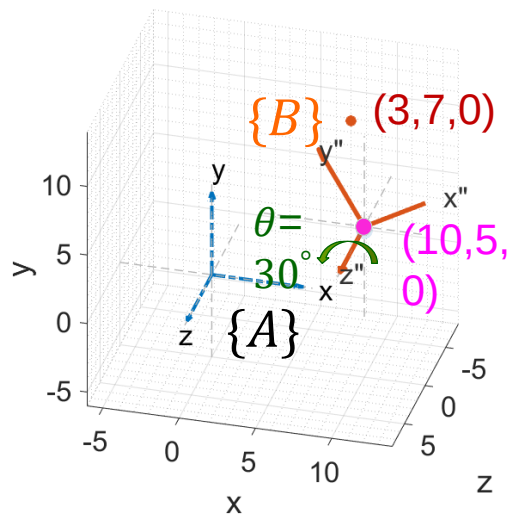


Mapping -3

□ Ex:

$${}^B P = \begin{bmatrix} 3 \\ 7 \\ 0 \end{bmatrix} \quad {}^A P_{Borg} = \begin{bmatrix} 10 \\ 5 \\ 0 \end{bmatrix} \quad {}^A \hat{X}_B = \begin{bmatrix} \frac{\sqrt{3}}{2} \\ \frac{1}{2} \\ \frac{1}{2} \\ 0 \end{bmatrix} \quad {}^A \hat{Y}_B = \begin{bmatrix} -\frac{1}{2} \\ \frac{\sqrt{3}}{2} \\ \frac{1}{2} \\ 0 \end{bmatrix} \quad {}^A \hat{Z}_B = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 1 \end{bmatrix} \Rightarrow {}^A P = ?$$

$$\begin{bmatrix} {}^A P \\ 1 \end{bmatrix} = \begin{bmatrix} {}^A R_B & {}^A P_{Borg} \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} {}^B P \\ 1 \end{bmatrix} = \underbrace{\begin{bmatrix} \frac{\sqrt{3}}{2} & -\frac{1}{2} & 0 & 10 \\ \frac{1}{2} & \frac{\sqrt{3}}{2} & 0 & 5 \\ \frac{1}{2} & \frac{1}{2} & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}}_{\text{Transformation Matrix}} \begin{bmatrix} 3 \\ 7 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 9.098 \\ 12.562 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} | \\ {}^A P \\ | \\ 1 \end{bmatrix}$$

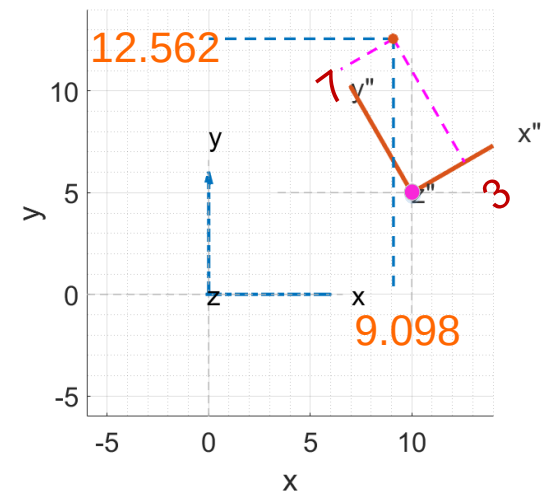


單純看 ${}^A T_B$: 表達{B} 相對於{A} 的方法

看整個操作：

轉換point在不同frame下的表達

投影至XY平面驗證答案



Operators -1

□ ${}^A_B T$ 除了 Mapping 之外，也可當 **Operator**，對向量（或點）

進行移動或轉動

◆ 僅有 移動

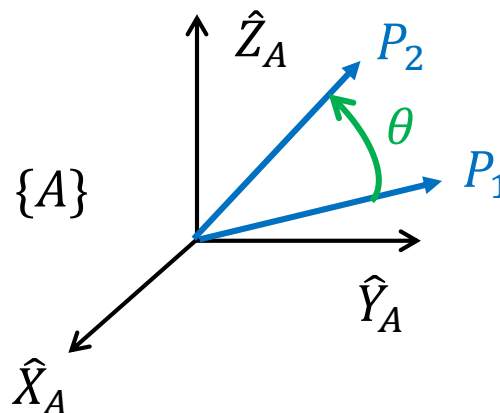
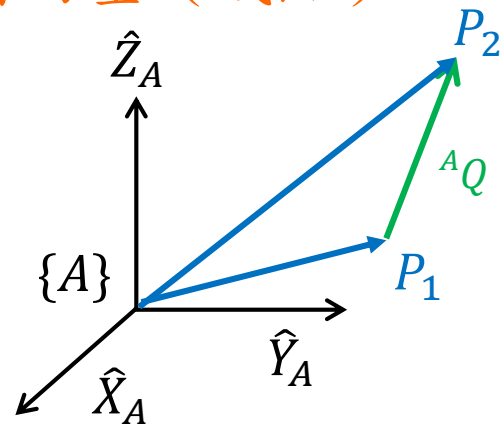
$${}^A P_2 = {}^A P_1 + {}^A Q$$

$$\begin{bmatrix} {}^A P_2 \\ 1 \end{bmatrix} = D(Q) \begin{bmatrix} {}^A P_1 \\ 1 \end{bmatrix} = \begin{bmatrix} I & {}^A Q \\ 0 & 1 \end{bmatrix} \begin{bmatrix} {}^A P_1 \\ 1 \end{bmatrix} = \begin{bmatrix} {}^A P_1 + {}^A Q \\ 1 \end{bmatrix}$$

◆ 僅有 轉動

$${}^A P_2 = R_{\hat{K}}(\theta) {}^A P_1$$

$$\begin{bmatrix} {}^A P_2 \\ 1 \end{bmatrix} = \begin{bmatrix} R_{\hat{K}}(\theta) & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} {}^A P_1 \\ 1 \end{bmatrix} = \begin{bmatrix} R_{\hat{K}}(\theta) {}^A P_1 \\ 1 \end{bmatrix}$$

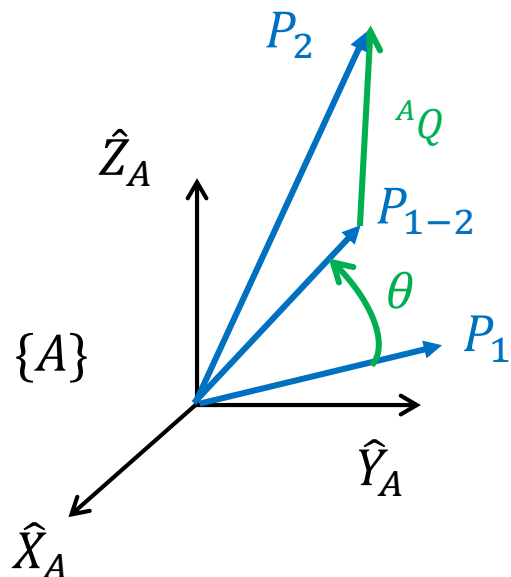


Operators -2

◆ 移動和轉動複合

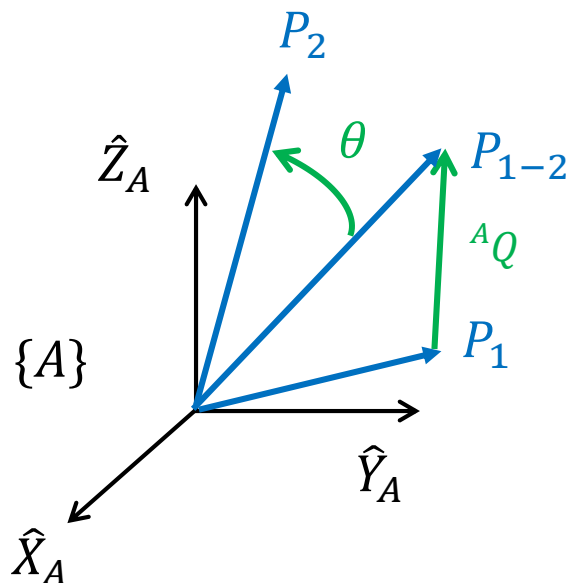
$${}^A P_{2\ 3 \times 1} = \underset{3 \times 3}{R_{\hat{K}}(\theta)} {}^A P_{1\ 3 \times 1} + {}^A Q_{3 \times 1} \quad \text{先轉動再移動}$$

$$\begin{bmatrix} {}^A P_2 \\ 1 \end{bmatrix} = \begin{bmatrix} R_{\hat{K}}(\theta) & {}^A Q \\ 0 & 1 \end{bmatrix} \begin{bmatrix} {}^A P_1 \\ 1 \end{bmatrix} = \begin{bmatrix} R_{\hat{K}}(\theta) {}^A P_1 + {}^A Q \\ 1 \end{bmatrix} = T \begin{bmatrix} {}^A P_1 \\ 1 \end{bmatrix}$$



先轉動再移動

$\neq$



先移動再轉動 (${}^A Q$ 也會被轉動到)

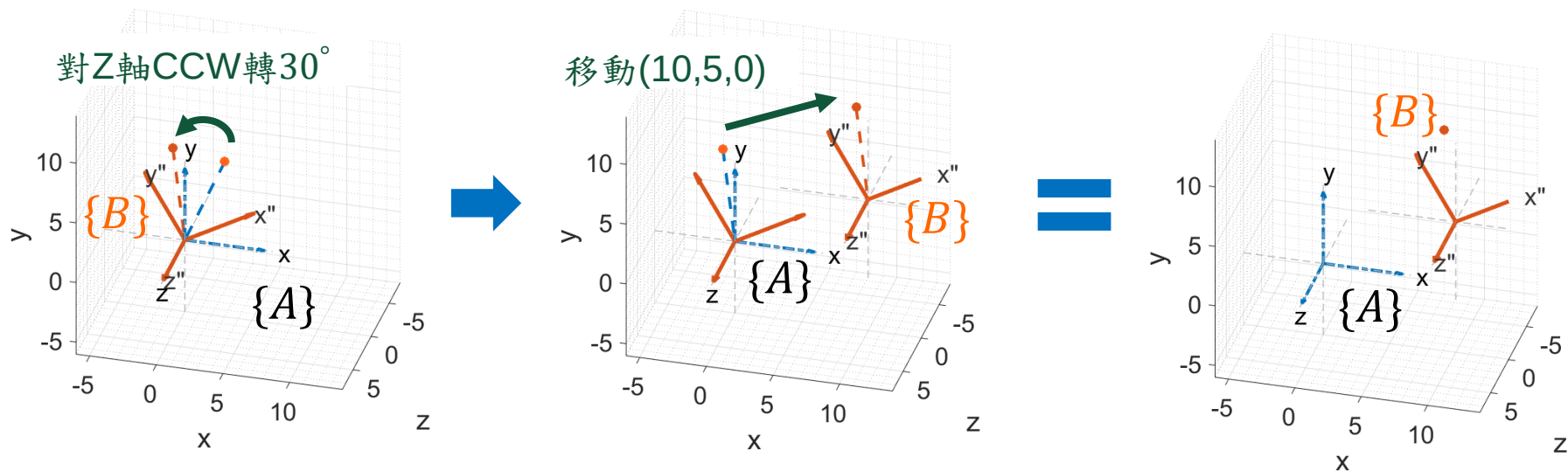
$${}^A P_2 = R_{\hat{K}}(\theta) ({}^A P_1 + {}^A Q) = R_{\hat{K}}(\theta) {}^A P_1 + R_{\hat{K}}(\theta) {}^A Q$$

Operators -3

□ EX: Point $P_1 = \begin{bmatrix} 3 \\ 7 \\ 0 \end{bmatrix}$, 先對Z軸CCW轉 30° , 然後移動 $\begin{bmatrix} 10 \\ 5 \\ 0 \end{bmatrix}$ 到 $P_2 \Rightarrow P_2 = ?$

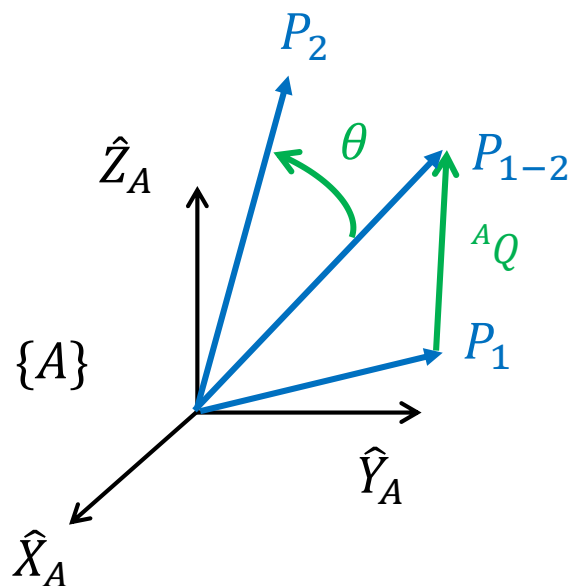
$$\begin{bmatrix} {}^A P_2 \\ 1 \end{bmatrix} = \begin{bmatrix} R_{\hat{K}}(\theta) & {}^A Q \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} {}^A P_1 \\ 1 \end{bmatrix} = \begin{bmatrix} \frac{\sqrt{3}}{2} & -\frac{1}{2} & 0 & 10 \\ 1 & \frac{\sqrt{3}}{2} & 0 & 5 \\ \frac{1}{2} & \frac{1}{2} & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 3 \\ 7 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 9.098 \\ 12.562 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} | \\ {}^A P_2 \\ | \\ 1 \end{bmatrix}$$

和「Mapping -3」的答案相同，Why?



Operators -4

- In-video Quiz: 如果要如下圖所示的先移動再轉動，那T應該如何表達？



A.
$$\begin{bmatrix} R_{\hat{K}}(\theta) & {}^A Q \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

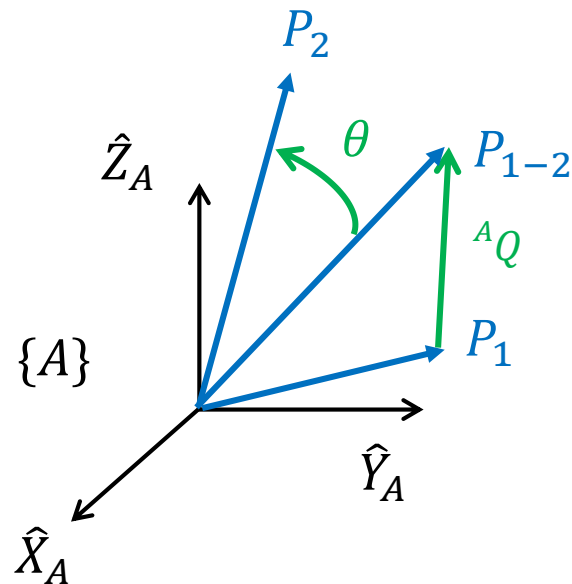
B.
$$\begin{bmatrix} I & {}^A Q \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} R_{\hat{K}}(\theta) & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

C.
$$\begin{bmatrix} R_{\hat{K}}(\theta) & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} I & {}^A Q \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

D.
$$\begin{bmatrix} R_{\hat{K}}(\theta) & {}^A Q \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} R_{\hat{K}}(\theta) & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Operators -4

- In-video Quiz: 如果要如下圖所示的先移動再轉動，那T應該如何表達？



A.
$$\begin{bmatrix} R_{\hat{K}}(\theta) & {}^A Q \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

B.
$$\begin{bmatrix} I & {}^A Q \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} R_{\hat{K}}(\theta) & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

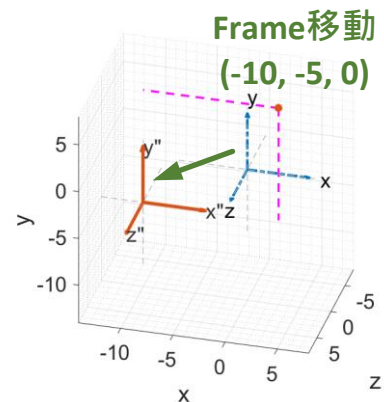
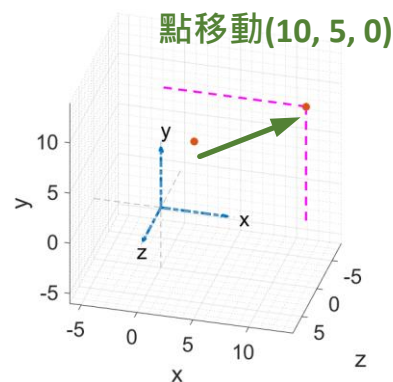
C.
$$\begin{bmatrix} R_{\hat{K}}(\theta) & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} I & {}^A Q \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

D.
$$\begin{bmatrix} R_{\hat{K}}(\theta) & {}^A Q \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} R_{\hat{K}}(\theta) & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

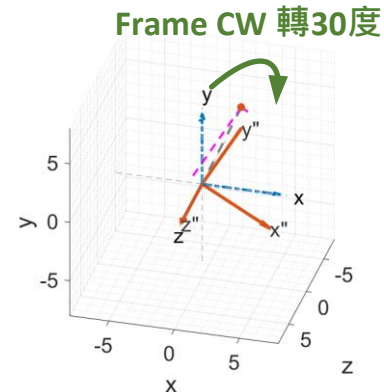
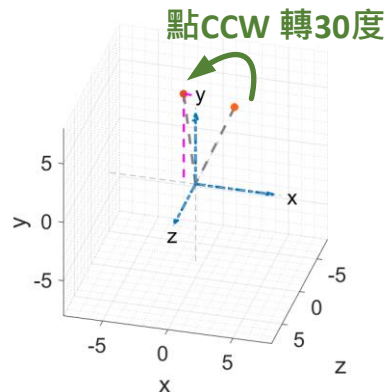
Operators -5

- 因為運動是相對的， A_BT 當Operator時對向量（或點）進行移動或轉動的操作，也可以想成是對frame進行「反向」的移動或轉動的操作

- ◆ Point往前移 = frame往後移



- ◆ Point逆時針轉 = frame順時針轉

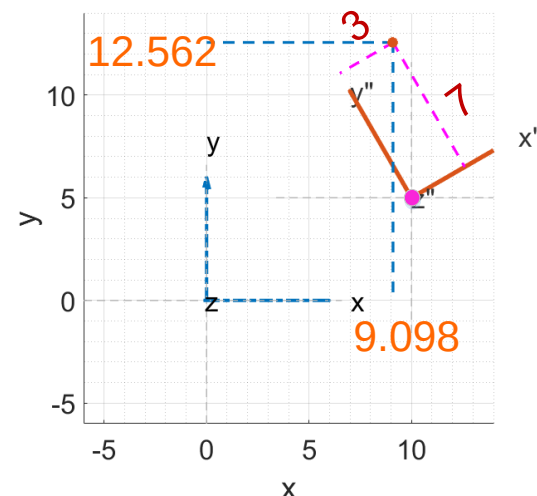
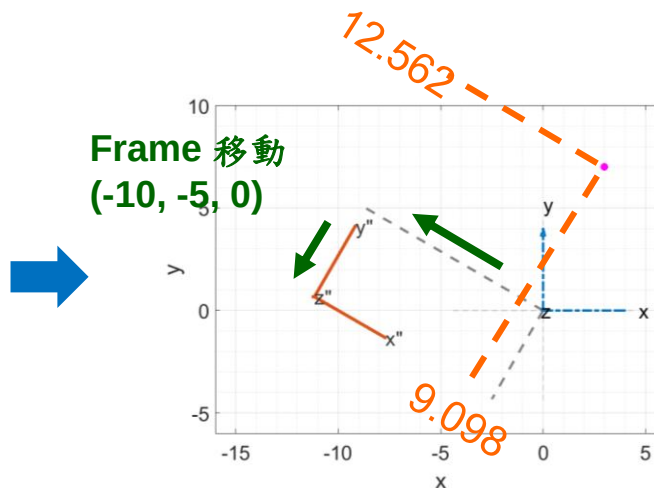
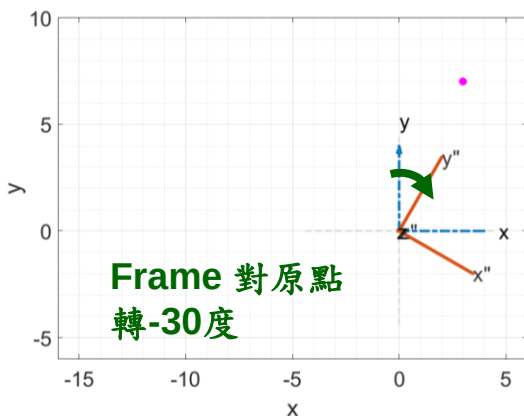


Operators -6

- Ex: Revisit Operator-3的範例，改以frame轉動的角度來想

Point $P_1 = \begin{bmatrix} 3 \\ 7 \\ 0 \end{bmatrix}$ ，先對Z軸CCW轉 30° ，然後移動 $\begin{bmatrix} 10 \\ 5 \\ 0 \end{bmatrix}$ 到 P_2 $\Rightarrow P_2 = ?$

投影至XY平面來看



與Operator -3的
答案相同

Transformation Matrix 小結

Homogeneous transformation matrix 的三種用法

- ◆ 描述一個frame(相對於另一個frame)的空間

狀態

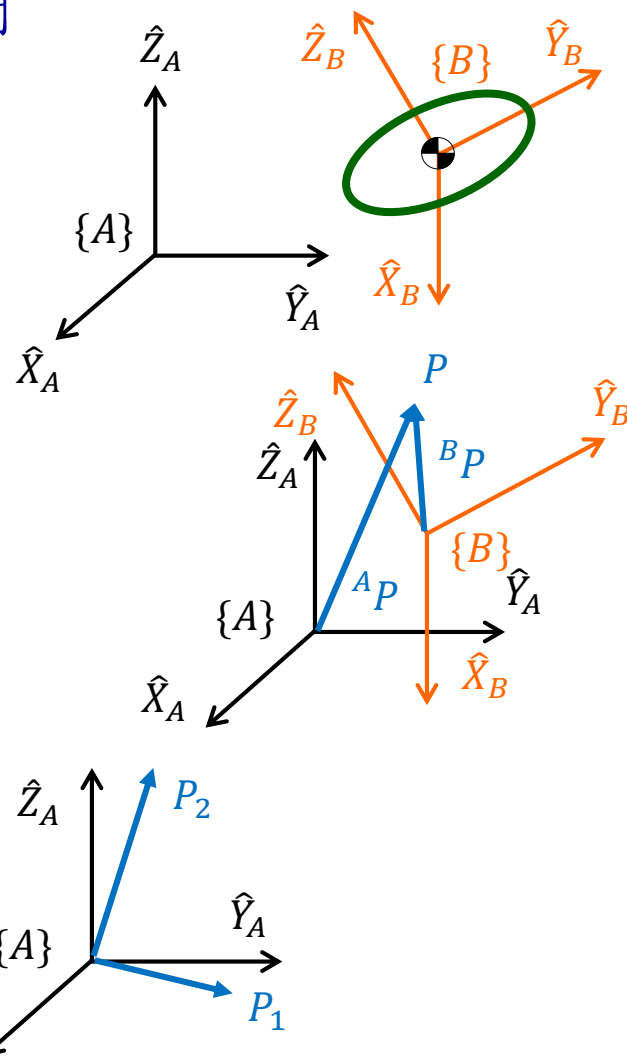
$${}^A_B T = \begin{bmatrix} | & | & | & | \\ {}^A\hat{X}_B & {}^A\hat{Y}_B & {}^A\hat{Z}_B & {}^A P_{B\text{ org}} \\ | & | & | & | \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

- ◆ 將point由某一個frame的表達換到另一個frame來表達

$$\begin{bmatrix} {}^A P \\ 1 \end{bmatrix} = {}^A_B T \begin{bmatrix} {}^B P \\ 1 \end{bmatrix}$$

- ◆ 將point(vector)在同一個frame中進行移動和轉動

$$\begin{bmatrix} {}^A P_2 \\ 1 \end{bmatrix} = T \begin{bmatrix} {}^A P_1 \\ 1 \end{bmatrix}$$





Transformation Matrix 運算 -1

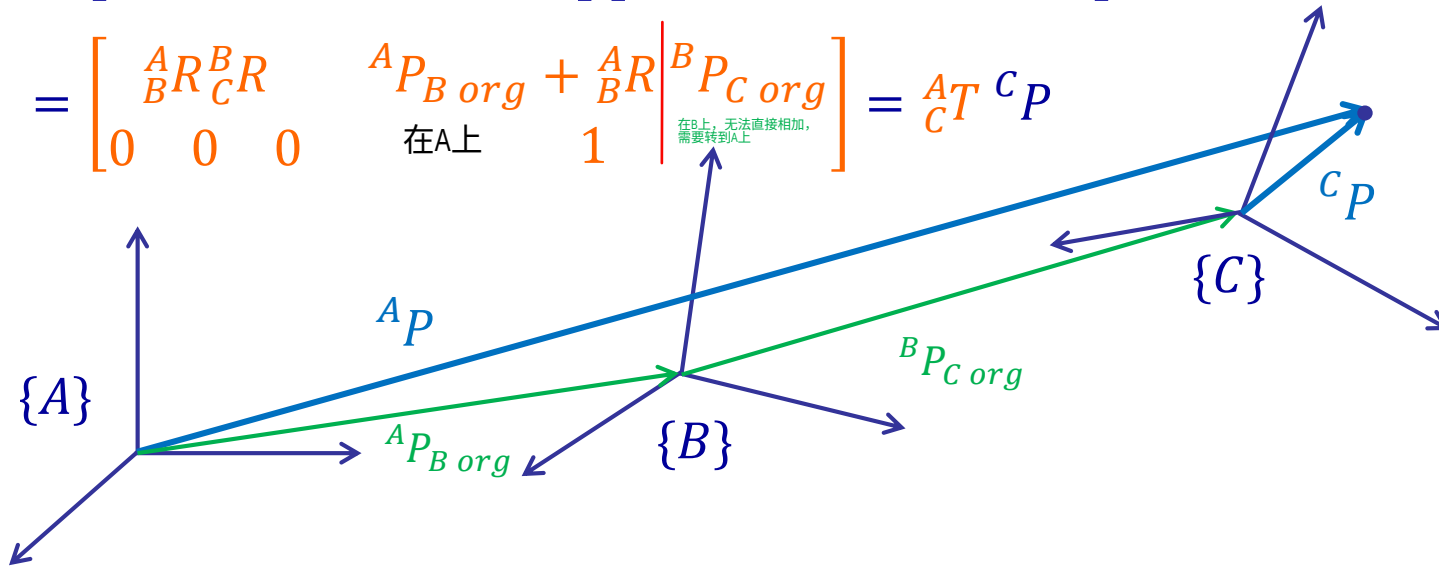
□ 連續運算

$${}^A P = {}^A_B T {}^B P = {}^A_B T ({}^B_C T {}^C P) = {}^A_B T {}^B_C T {}^C P$$

$$= \left[\begin{array}{ccc|c} {}^A_B R & & & {}^A P_{B \text{ org}} \\ \hline 0 & 0 & 0 & 1 \end{array} \right] \left[\begin{array}{ccc|c} {}^B_C R & & & {}^B P_{C \text{ org}} \\ \hline 0 & 0 & 0 & 1 \end{array} \right] {}^C P$$

$$= \left[\begin{array}{ccc|c} {}^A_B R {}^B_C R & {}^A P_{B \text{ org}} + {}^A_B R {}^B P_{C \text{ org}} & & 1 \\ \hline 0 & 0 & 0 & 1 \end{array} \right] {}^C P$$

在B上，无法直接相加，
需要转到A上



$${}^A P = {}^A_B T {}^B_C T {}^C_D T {}^D P$$

$$= \left[\begin{array}{ccc|c} {}^A_B R {}^B_C R {}^C_D R & {}^A P_{B \text{ org}} + {}^A_B R {}^B P_{C \text{ org}} + {}^A_B R {}^B_C R {}^C P_{D \text{ org}} & & 1 \\ \hline 0 & 0 & 0 & 1 \end{array} \right] {}^D P$$

Transformation Matrix 運算 -2

□ 反矩陣 ${}^A_B T = \begin{bmatrix} {}^A_B R & {}^A P_{B\text{org}} \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad {}^B_A T = {}^A_B T^{-1} = ?$

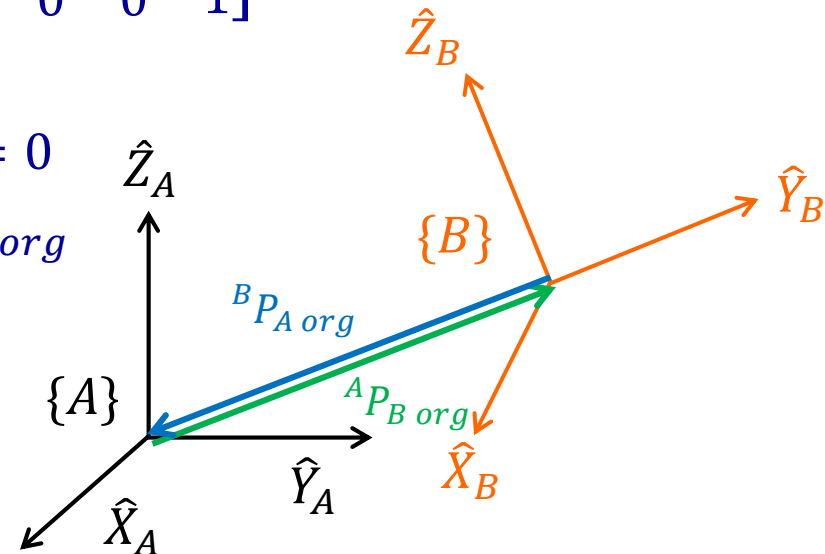
逆矩陣：逆過程

$$\begin{aligned} {}^A_B T {}^B_A T &= {}^A_B T {}^A_B T^{-1} = I_{4 \times 4} \\ \begin{bmatrix} {}^A_B R & {}^A P_{B\text{org}} \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} {}^B_A R & {}^B P_{A\text{org}} \\ 0 & 0 & 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} {}^A_B R {}^B_A R & {}^A P_{B\text{org}} + {}^A_B R {}^B P_{A\text{org}} \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} I_{3 \times 3} & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \end{aligned}$$

$$\begin{aligned} {}^A_B R {}^B_A R &= I_{3 \times 3} \\ \Rightarrow {}^B_A R &= {}^A_B R^T \end{aligned}$$

$$\begin{aligned} {}^A P_{B\text{org}} + {}^A_B R {}^B P_{A\text{org}} &= 0 \\ \Rightarrow {}^B P_{A\text{org}} &= -{}^A_B R^T {}^A P_{B\text{org}} \end{aligned}$$

$$\Rightarrow {}^A_B T^{-1} = \begin{bmatrix} {}^A_B R^T & -{}^A_B R^T {}^A P_{B\text{org}} \\ 0 & 0 & 0 & 1 \end{bmatrix}$$



Transformation Matrix 運算 -3

□ 連續運算，求未知之相對關係

$${}^U_D T = {}^U_A T {}^A_D T = {}^U_B T {}^B_C T {}^C_D T$$

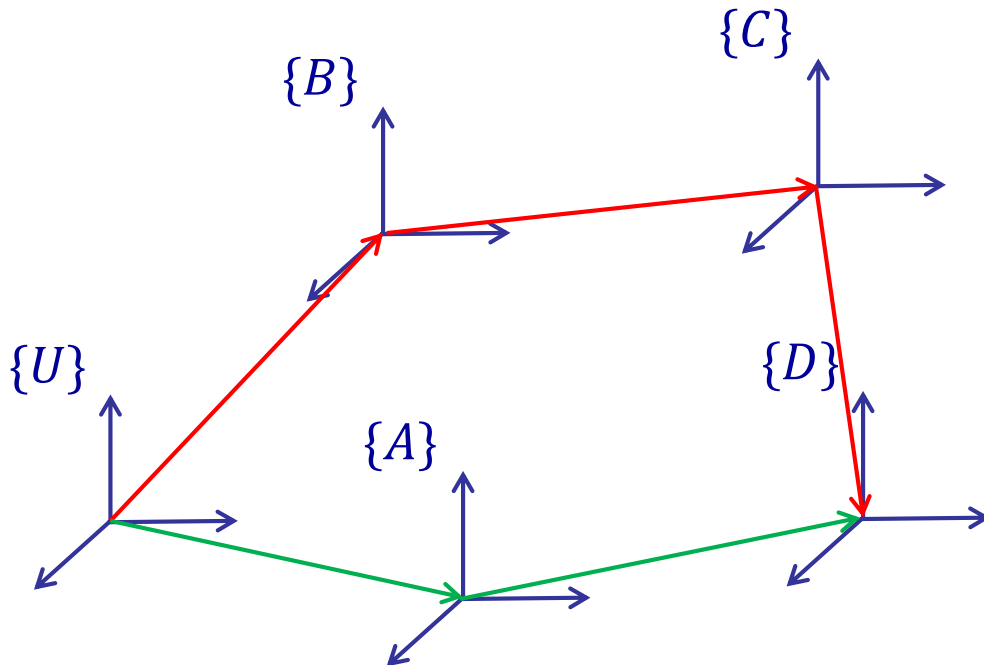
if ${}^C_D T$ unknown

$$= ({}^U_B T {}^B_C T)^{-1} {}^U_A T {}^A_D T$$

$$= {}^B_C T^{-1} {}^U_B T^{-1} {}^U_A T {}^A_D T$$

if ${}^B_C T$ unknown

$$= {}^U_B T^{-1} {}^U_A T {}^A_D T {}^C_D T^{-1}$$



Transformation Matrix 運算 -4

□ 連續運算 法則

- ◆ Initial condition: $\{A\}$ and $\{B\}$ coincide ${}^A_B T = I_{4 \times 4}$
- ◆ $\{B\}$ 對 $\{A\}$ 的轉軸旋轉：用 “premultiply” 左乘：每次变换相对固定的 $\{A\}$ 进行

- 以operator來想，對某一個向量，「以同一個座標為基準」，進行轉動或移動的操作
- Ex: $\{B\}$ 依序經過 T_1 、 T_2 、 T_3 三次transformations

$${}^A_B T = T_3 T_2 T_1 I \quad v' = {}^A_B T v = T_3 T_2 T_1 v$$

- ◆ $\{B\}$ 對 $\{B\}$ 自身的轉軸旋轉：用 “postmultiply” 右乘：每次变换相对物体自己坐标轴进行
- 以mapping來想，對某一個向量，從最後一個frame「逐漸轉動或移動」來回到第一個frame
- Ex: $\{B\}$ 依序經過 T_1 、 T_2 、 T_3 三次transformation

$${}^A_B T = I T_1 T_2 T_3 \quad {}^A P = {}^A_B T {}^B P = I T_1 T_2 T_3 {}^B P$$



Transformation Matrix 運算 -5

□ 連續運算 小結

- ◆ 以固定的{A}或移動的{B}為基準進行移動換轉動操作，
transformation matrix應用不同的連乘方式
- ◆ 思考邏輯和考量Fixed angles vs. Euler angles的連續旋轉順序相似